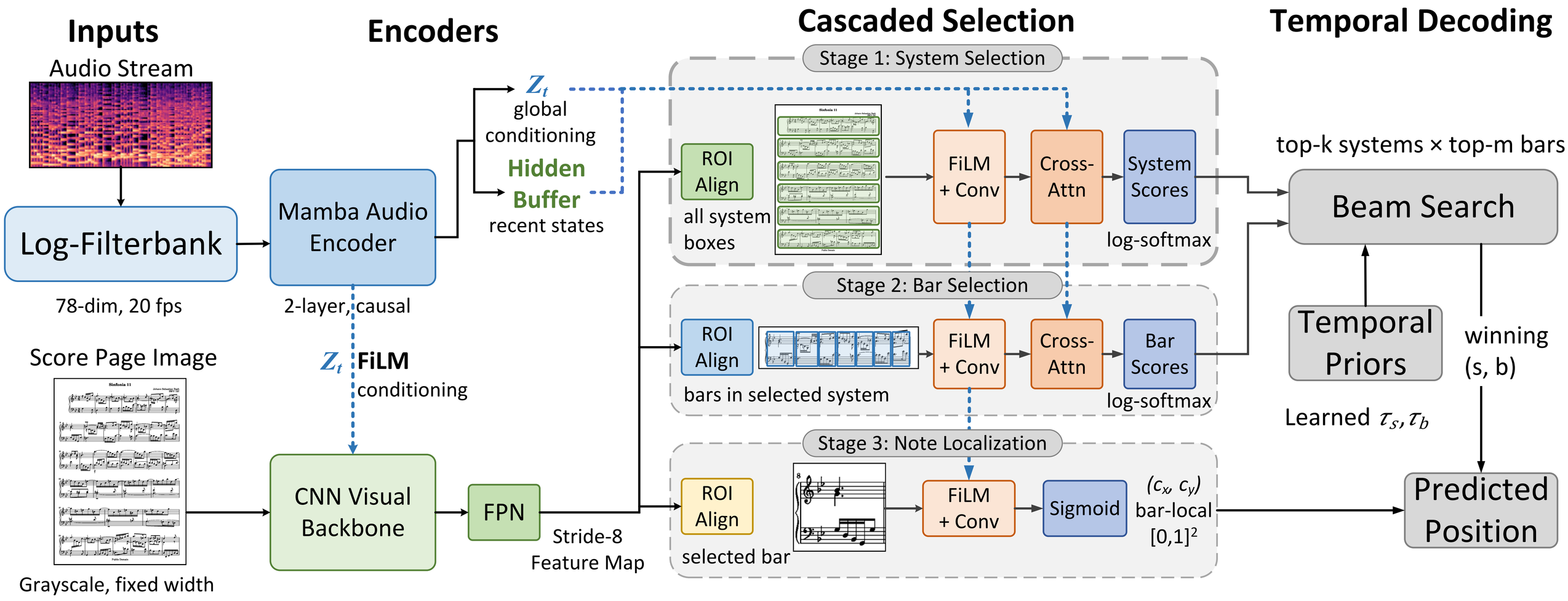
Inputs
Encoders
Cascaded Selection
Temporal Decoding
Audio Stream
Log-Filterbank
78-dim, 20 fps
Mamba Audio Encoder
2-layer, causal
Z_t
global conditioning
Hidden Buffer
recent states
Score Page Image
Grayscale, fixed width
Z_t FiLM conditioning
CNN Visual Backbone
FPN
Stride-8 Feature Map
Stage 1: System Selection
ROI Align
all system boxes
FiLM + Conv
Cross-Attn
System Scores
log-softmax
Stage 2: Bar Selection
ROI Align
bars in selected system
FiLM + Conv
Cross-Attn
Bar Scores
log-softmax
Stage 3: Note Localization
ROI Align
selected bar
FiLM + Conv
Sigmoid
(c_x, c_y)
bar-local
$[0,1]^2$
top-k systems × top-m bars
Beam Search
Temporal Priors
Learned τ_s, τ_b
winning (s, b)
Predicted Position